

Heart Study Final Project

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Data Set: Framingham Heart Study

- Large long-term study with data collected from patient examinations over the course of several decades
- Data is collected one-time at examination, not over time
- Contains 16 columns: 'male', 'age', 'education', 'currentSmoker', 'cigsPerDay', 'BPMeds', 'prevalentStroke', 'prevalentHyp', 'diabetes', 'totChol', 'sysBP', 'diaBP', 'BMI', 'heartRate', 'glucose', 'TenYearCHD'
- Has 4240 rows
- Found on Kaggle



Data Preparation

- Data had a total of 645 null values spread across the columns, all float64 data types
- Used the mean of the columns to fill in null vales
- Used interquartile method to detect and remove outliers, leaving 3659 rows

	lower bound	upper bound	number of outliers	percentage of outliers
education	-2	6	0	0
cigsPerDay	-30	50	12	0.283
totChol	122	346	57	1.3443
sysBP	76.5	184.5	126	2.9717
diaBP	52.5	112.5	77	1.816
BMI	15.645	35.465	97	2.2877
heartRate	45.5	105.5	76	1.7925
glucose	52.5	104.5	262	6.1792



EDA - Summary Statistics

	male	age	education	currentSmoker	cigsPerDay	BPMeds	prevalentStroke	prevalentHyp	diabetes	totChol	sysBP	diaBP	BMI	heartRate	glucose	TenYearCHD
count	3659	3659	3659	3659	3659	3659	3659	3659	3659	3659	3659	3659	3659	3659	3659	3659
mean	0.4378	49.0522	2.0126	0.5018	9.0951	0.0214	0.0049	0.2648	0.0052	234.1159	129.0249	81.5496	25.3989	74.7581	78.2958	0.1339
std	0.4962	8.4783	1.0089	0.5001	11.6594	0.1435	0.07	0.4413	0.0719	40.2596	18.0668	10.402	3.4747	10.9006	9.887	0.3406
min	0	32	1	0	0	0	0	0	0	124	83.5	53	16.48	46	53	0
25%	0	42	1	0	0	0	0	0	0	205	116	74	22.935	67	72	0
50%	0	48	2	1	1	0	0	0	0	232	126.5	81	25.2	75	78	0
75%	1	56	3	1	20	0	0	1	0	260	140	88	27.72	80	84	0
max	1	70	4	1	50	1	1	1	1	346	184.5	112.5	35.45	105	104	1

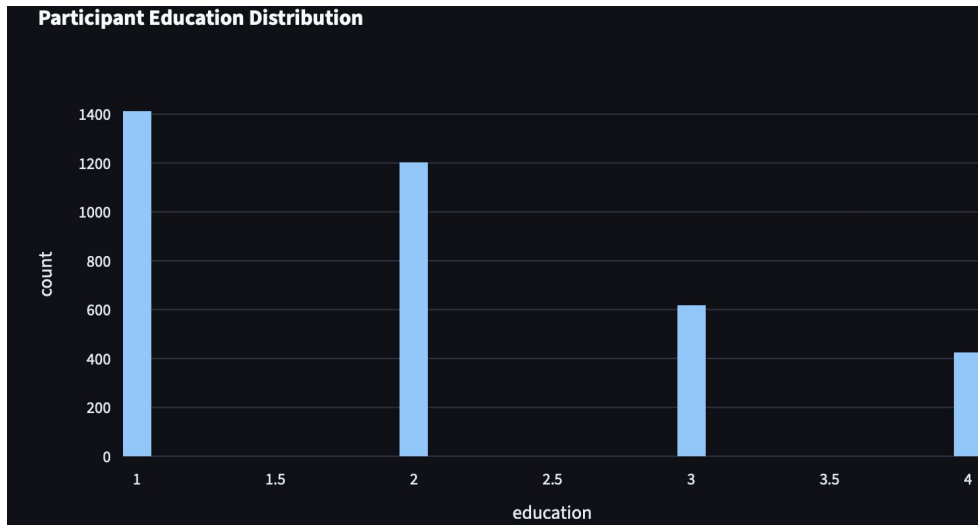


Summary Statistics Cont'd

- Age, cigsPerDay, totChol, sysBP, diaBP, heartRate, and glucose all have a higher std dev
- Diaetes, Prevalent stroke, and BPMeds all have a very low mean
- These are binary values, either 0 or 1, meaning a very large portion of the participants register as 0 for these columns
 - Meaning for instance, most do not have diabetes
- Current Smoker, another yes or no column, has a mean of about 0.5, meaning close to half of the participants are current smokers and half are not
- Participant ages range from 32 to 70 years old, with a mean of about 49 years old



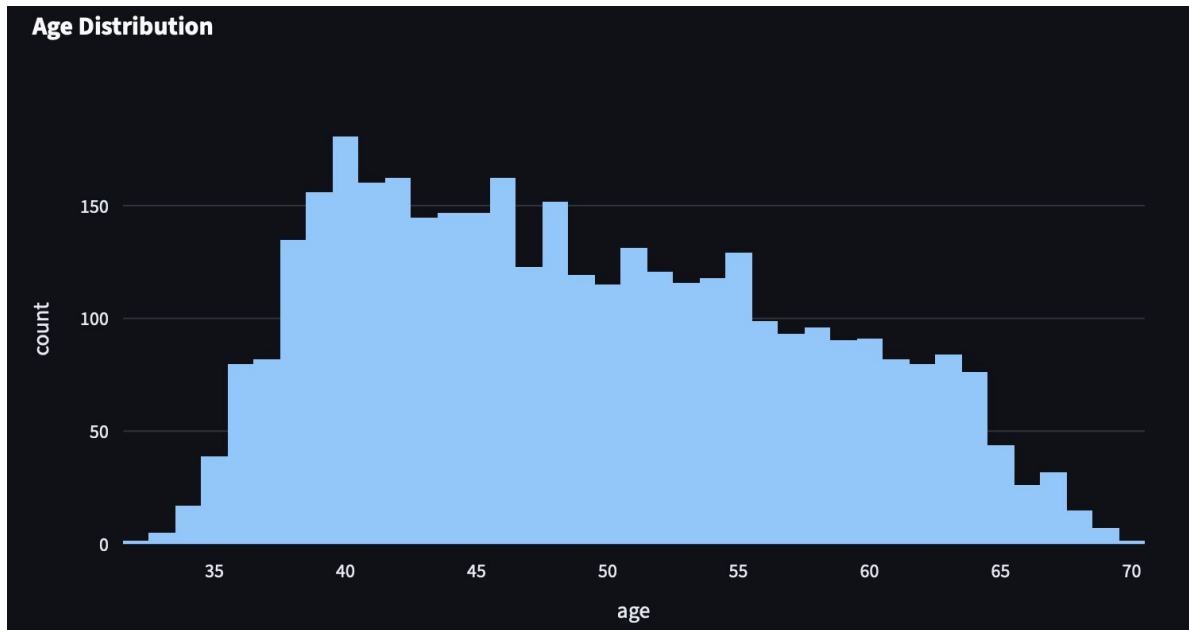
Data Visualization



- The higher the education level, the lower number of participants in that threshold

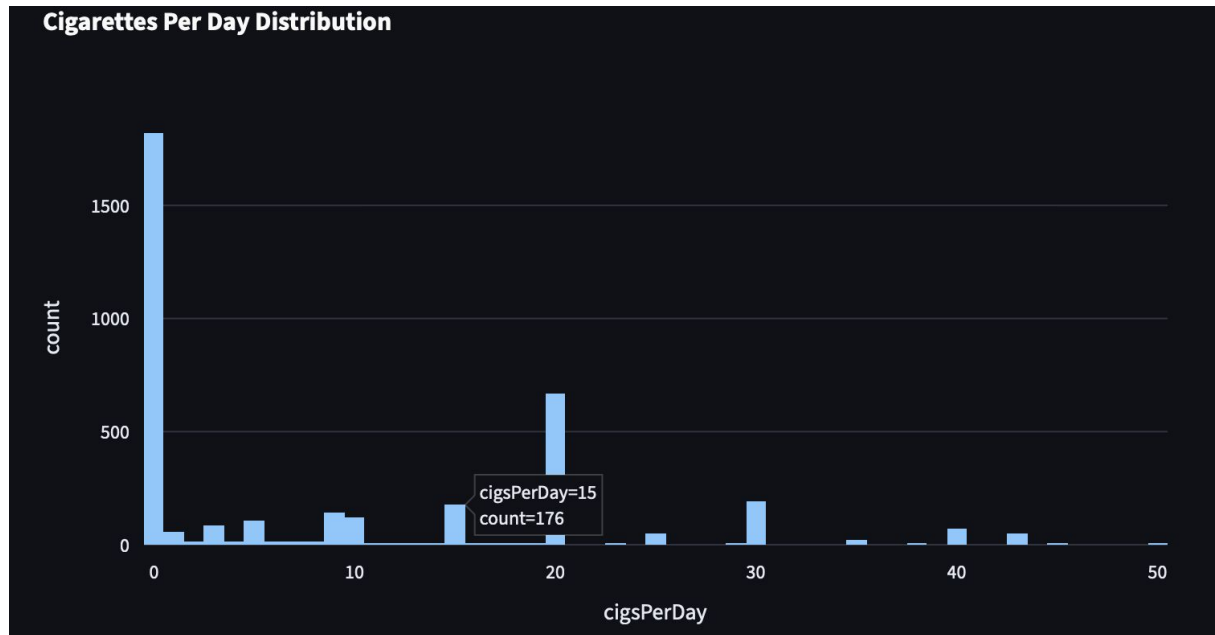


Data Visualization Cont'd



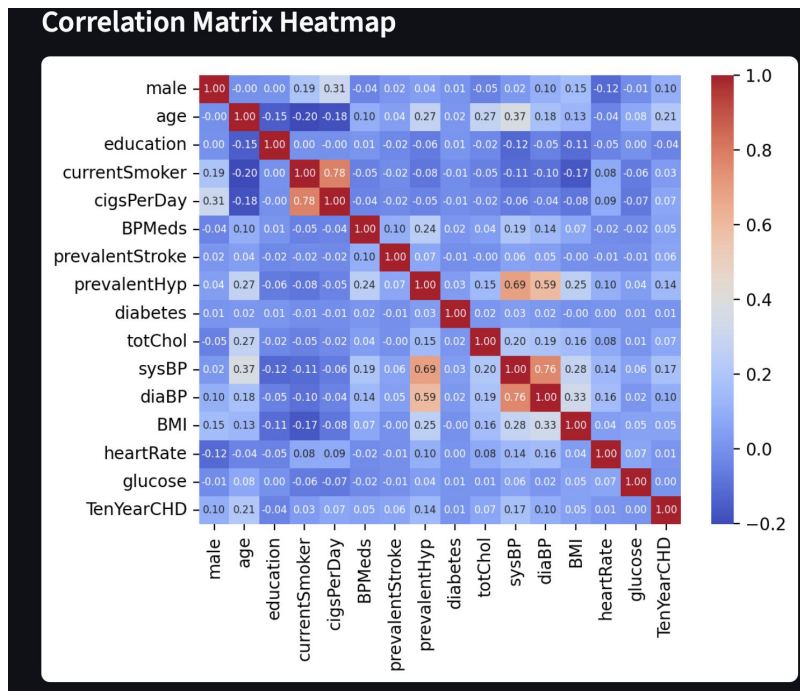


Data Visualization Cont'd





Correlation Matrix



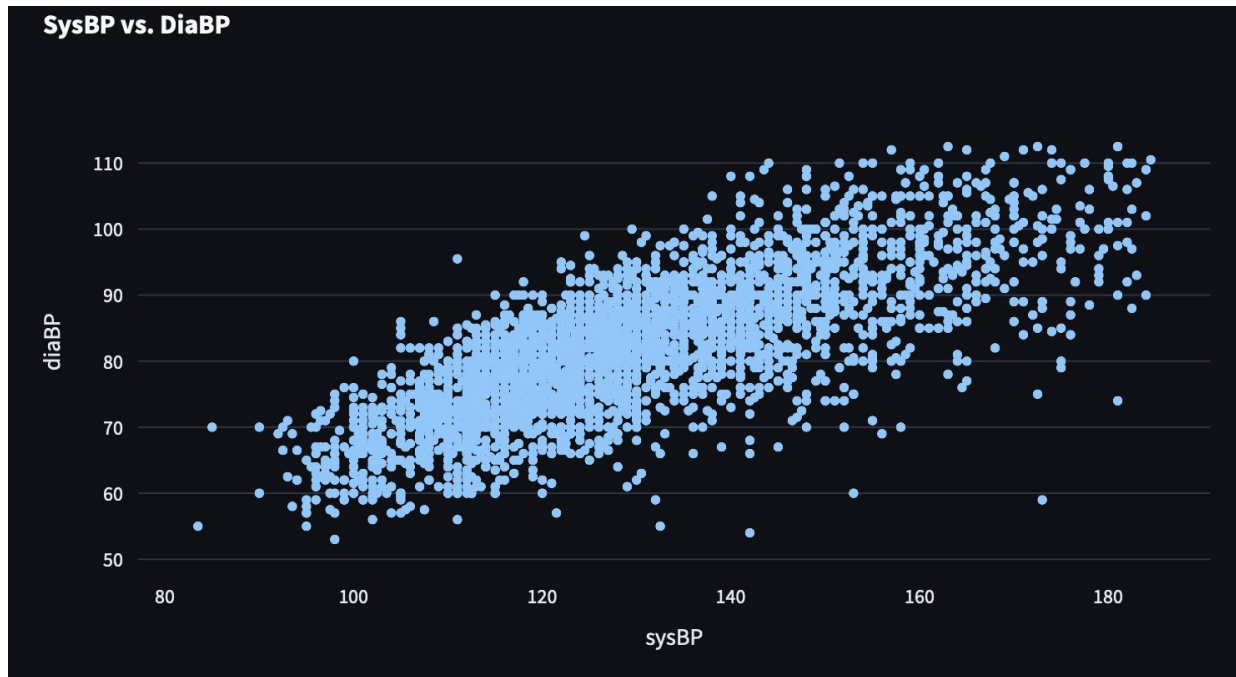


Correlation Matrix Cont'd

- sysBP and diaBP are highly correlated
- sysBP and diaBP are both highly correlated with prevalentHyp
- cigsPerDay and current Smoker are highly correlated
- Some correlation between diaBP and BMI, sysBP and BMI, and prevalentHyp and BMI
- Some correlation between BPMeds and prevalentHyp
 - Makes sense as those taking meds for blood pressure might have blood flow issues
- Decent correlation between age and prevalentHyp, age and totChol, and age and sysBP

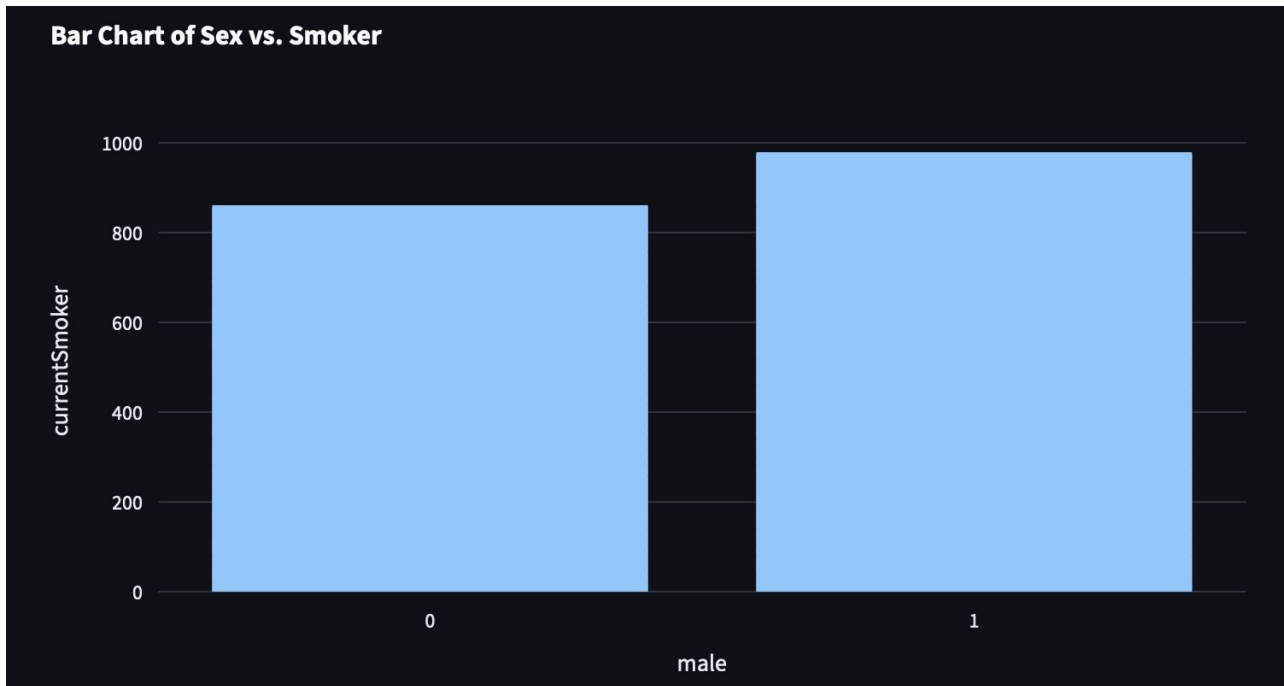


Data Visualization Cont'd





Data Visualization Cont'd





Hypothesis Testing

Does the effect of smoking on systolic blood pressure differ between males and females?

Outcome (Dependent Variable):

- Systolic Blood Pressure (sysBP)

Primary Predictor (Independent Variable):

- Smoking status (smoker = 1, non-smoker = 0)
- currentSmoker

Effect Modifier:

- Gender (male = 1, female = 0)



Hypothesis Testing

H0 : The relationship between smoking and systolic BP is the same across genders

H1 : The relationship between smoking and systolic BP differs by gender

	Group	T-Statistic	P-Value
0	Males	-2.0079	0.0449
1	Females	-7.6326	0.00000000000003

After conducting the t-test, males have a t-score of -2.0079 and p-value of 0.0449, and females have a t-score of -7.6326 and p-value of <0.0001 . Since the p-value of both genders is < 0.05 , they are both statistically significant and we fail to reject the null hypothesis (H0). Smoking significantly affects systolic blood pressure in both males and females, but not in the way we expected.



Hypothesis Testing

Upon closer inspection, the mean BP of smokers is actually LOWER than non-smokers in both genders, which is the opposite of general expectations. This points to the presence of confounding factors such as age or BMI (BP tends to increase with age / have a higher BMI).

Future studies:

- Adjust for confounders to isolate smoking and systolic BP from age, cholesterol and BMI.
- ANOVA two-way test on the interaction of smoking and gender on BP.



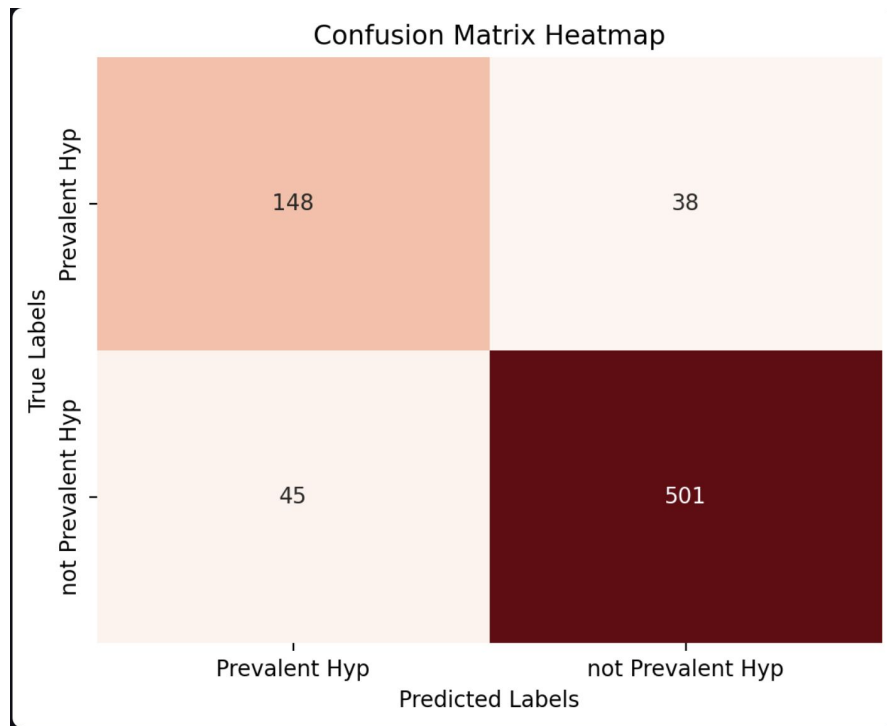
Machine Learning

- Using the data to predict Prevalent Hypertension
- Supervised Learning problem
 - Binary classification labels: prevalentHyp = 1,
no prevalentHyp = 0
- Using RandomForestClassifier of SciKitLearn
- Split Training and Testing into 80/20 split
- Used StandardScaler for input features



Machine Learning Cont'd

- Accuracy: 88.66%
- F1 Score: 0.78





Machine Learning Cont'd

- Initially trained using only one input feature at a time, with very low accuracy scores
- Gradually increased to many usable input features and obtained reasonably high accuracy scores
- Had to specify `class_weight='balanced'` for best results in classifier instantiation
- Results agree with intuition as many of the features in the data set seem to be contributing factors or correlated with heart disease and hypertension, such as smoking and age



Conclusion

- Chose to replace null values with the column means, could see how using another method such as median, zeroes, or deletion affects the analysis
- Used Inter Quartile Method discussed in class to detect outliers and decided to remove them instead of replacing them
- RandomForestClassifier worked well and had a decent accuracy
 - Data set includes many risk factors correlated with heart disease so it makes sense that using these features as input would lead to predicted prevalent hypertension



Future Considerations

- Could try other supervised learning classification models besides random forest, such as Naive Bayes or support vector machines
- It is important to remember the big picture when using long-term observational data. Longitudinal studies over time are a better match for the hypothesis tested
- More focused studies and datasets would have been more useful in the tests we conducted. There was a lot of unused variables in our dataset.
- Future tests should adjust for confounding variables. The tested variables should be isolated and tested for interactions. An ANOVA two-way test may be better suited for our hypothesis.